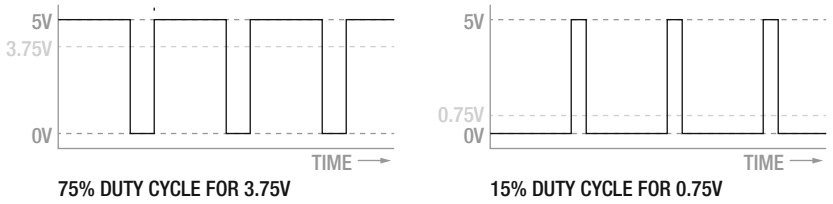

Figure 6-2. *Pulse width modulation duty cycle*

Figure 6-2 shows how a PWM signal that is on, or +5v, for 75% of the time and off, or 0v, for 25% will simulate an analog voltage of +3.75v. This would be referred to as a 75% duty cycle. Likewise, a duty cycle of 15%, where the pin is on for 15% of the time and off 85% of the time, would result in only an average of +0.75v. This is how we can use PWM to give an LED the appearance of fading or with the right hardware to change the speed of a motor. Now that we have a handle on how analog works on the Arduino, let's look at our next project.

Project 5: Telematic Breath

In our fifth project, we will sense and re-create or "broadcast" a breath or gentle breeze from one location to another using a simple wind sensor and a small fan. Breathe on the sensor and the fan will spin at a speed that correlates to the force of your breath. In this way, applying the data from an analog sensor to the speed of a fan will provide a very tangible experience of transmitting data from one source to another.

Hooking It Up

While the hardware is a little more complex than previous projects, it's not really all that bad, with a total of four components. We have specified a very particular sensor and fan at the beginning of this chapter to complete this project, although just about any analog sensor could be used in place of the wind sensor and about any other DC fan, motor, solenoid, or other actuator could be used to substitute the fan. Likewise you might change out the transistor for another switching device like a relay or MOSFET, detailed later in Chapters 11 and 12.

This particular wind sensor is a low-cost thermal anemometer developed by Paul Badger at Modern Device to measure wind speed. Specifically, it measures the electricity needed to maintain a specific temperature on a heating element as the wind changes. This sensor is remarkably sensitive and can detect the slightest breath. We will couple this with a fan to re-create the wind speed input by controlling the speed of the fan through PWM.

The fan we chose will run on +5v DC, but it exceeds the current available to the Arduino's output pins, so to power it safely, we need to use a small transistor to switch the bigger load. The transistor we are using is the 2N3904, although the 2N2222 is pin compatible and you could even substitute any of a number of other similar devices. The 2N3904 has 3 pins labeled e, b, and c—for emitter, base, and collector respectively. These names refer to what each pin does: emitter is connected to ground; collector to the negative side of the load, in this case the black wire on the fan; and base connects to a series resistor and to the PWM pin on the Arduino.